

RNN

- Replaced TDSConvEncoder with RNNEncoder
- Running on VS Code and Google Colab.

	VS Code 1	VS Code 2	Colab 1	Colab 2	Colab 3
num_parameters	3.7M	3.7M	3.7M	7.3M	3.7M
hidden_size	384	384	384	512	384
num_layers	3	3	3	4	3
dropout	0.2	0.2	0.2	0.3	0.2
optimizer.lr	0.0001	0.0001	0.001	0.001	0.001
scheduler	Constant LR	Constant LR	LinearWarmupCosineAnnealingLR	LinearWarmupCosineAnnealingLR	LinearWarmupCosineAnnealingLR
warmup_epochs	—	—	10	10	10
max_epochs	40	80	100, extending to 150, then 200	150	200
val CER	57.709	41.83	38.17	50.22	40.56
test CER	—	38.75	31.29	38.92	33.24

VS Code Run 1:

RNN on VS Code after 40 epochs:

Validate metric DataLoader 0

```
-----
val/CER      57.7093505859375
val/DER      4.342046737670898
val/IER      30.17279624938965
val/SER      23.19450569152832
val/loss     1.6620607376098633
```

VS Code Run 2:

Same set up but for 80 (unfortunately, couldn't restart from checkpoint).

	Name	Type	Params
0	model	Sequential	3.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

3.7 M	Trainable params		

```
module:
  _target_: emg2qwerty.lightning.RNNCTCModule
  in_features: 528
  mlp_features:
    - 384
  hidden_size: 384
  num_layers: 3
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.0001
lr_scheduler:
  scheduler:
    _target_: torch.optim.lr_scheduler.ConstantLR
    factor: 1.0
    total_iters: 80
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: cpu
  devices: 1
```

```
num_nodes: 1
max_epochs: 80
```

```
-m emg2qwerty.train
user="single_user"
dataset.root=./data/89335547
trainer.accelerator=cpu
trainer.devices=1
trainer.max_epochs=80
batch_size=32
num_workers=4
module._target_=emg2qwerty.lightning.RNNCTCModule
~module.block_channels
~module.kernel_width
+module.hidden_size=384
+module.num_layers=3
+module.dropout=0.2
~cluster optimizer.lr=0.0001
lr_scheduler.scheduler._target_=torch.optim.lr_scheduler.ConstantLR
+lr_scheduler.scheduler.factor=1.0
+lr_scheduler.scheduler.total_iters=80
~lr_scheduler.scheduler.T_max
~lr_scheduler.scheduler.eta_min
```

Colab Run 1:

	Name	Type	Params
0	model	Sequential	3.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

3.7 M	Trainable params		
0	Non-trainable params		
3.7 M	Total params		
14.934	Total estimated model params size (MB)		

```
module:
  _target_: emg2qwerty.lightning.RNNCTCModule
  in_features: 528
  mlp_features:
    - 384
```

```

hidden_size: 384
num_layers: 3
dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 100

```

	Name	Type	Params
0	model	Sequential	3.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
3.7 M	Trainable params		
0	Non-trainable params		
3.7 M	Total params		
14.934	Total estimated model params size (MB)		

Runningstage.validating metric	DataLoader 0
val/CER	39.12273025512695
val/DER	7.642889022827148
val/IER	11.962782859802246
val/SER	19.517057418823242
val/loss	1.2221662998199463

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.51s/it]

Runningstage.testing metric	DataLoader 0
test/CER	30.840717315673828
test/DER	2.7015345096588135
test/IER	6.24594783782959
test/SER	21.89323616027832
test/loss	0.9937214851379395

```
{'val_metrics': [{'val/loss': 1.2221662998199463,
  'val/CER': 39.12273025512695,
  'val/IER': 11.962782859802246,
  'val/DER': 7.642889022827148,
  'val/SER': 19.517057418823242}],
'test_metrics': [{'test/loss': 0.9937214851379395,
  'test/CER': 30.840717315673828,
  'test/IER': 6.24594783782959,
  'test/DER': 2.7015345096588135,
  'test/SER': 21.89323616027832}],
'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-11/08-30-29/checkpoints/epoch=92-step=11160.ckpt'}
```

Accidentally didn't restart from the checkpoint at Epoch 92 and ran for 58 epochs from scratch:

Runningstage.validating metric	DataLoader 0
val/CER	44.240142822265625
val/DER	9.282233238220215
val/IER	12.738147735595703
val/SER	22.21976089477539
val/loss	1.3778818845748901

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.49s/it]

Runningstage.testing metric	DataLoader 0
test/CER	36.39507293701172
test/DER	2.182839870452881
test/IER	9.811973571777344
test/SER	24.400259017944336
test/loss	1.1464358568191528

```
{'val_metrics': [{'val/loss': 1.3778818845748901,
  'val/CER': 44.240142822265625,
  'val/IER': 12.738147735595703,
  'val/DER': 9.282233238220215,
  'val/SER': 22.21976089477539}],
'test_metrics': [{'test/loss': 1.1464358568191528,
  'test/CER': 36.39507293701172,
  'test/IER': 9.811973571777344,
  'test/DER': 2.182839870452881,
  'test/SER': 24.400259017944336}],
'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-11/16-41-30/checkpoints/epoch=57-step=6960.ckpt'}
```

Added 50 epochs to the original 100:

Runningstage.validating metric	DataLoader 0
val/CER	39.277801513671875
val/DER	6.6681437492370605
val/IER	13.513513565063477
val/SER	19.096145629882812
val/loss	1.2291899919509888

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.48s/it]

Runningstage.testing metric	DataLoader 0
test/CER	30.602981567382812
test/DER	1.8586556911468506
test/IER	6.526907444000244
test/SER	22.217418670654297
test/loss	0.9875145554542542

```
{'val_metrics': [{'val/loss': 1.2291899919509888,
                  'val/CER': 39.277801513671875,
                  'val/IER': 13.513513565063477,
                  'val/DER': 6.6681437492370605,
                  'val/SER': 19.096145629882812}],
 'test_metrics': [{'test/loss': 0.9875145554542542,
                   'test/CER': 30.602981567382812,
                   'test/IER': 6.526907444000244,
                   'test/DER': 1.8586556911468506,
                   'test/SER': 22.217418670654297}],
 'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-11/rnn-resume/checkpoints/epoch=115-step=13920.ckpt'}
```

Added another 50 epochs for 200 total:

Runningstage.validating metric	DataLoader 0
val/CER	38.170135498046875
val/DER	6.557376861572266
val/IER	12.715994834899902
val/SER	18.896764755249023
val/loss	1.2042109966278076

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.49s/it]

Runningstage.testing metric	DataLoader 0
test/CER	31.294574737548828
test/DER	2.766371250152588
test/IER	5.575967311859131
test/SER	22.95223617553711
test/loss	0.9963287711143494

```
{'val_metrics': [{'val/loss': 1.2042109966278076,
                  'val/CER': 38.170135498046875,
                  'val/IER': 12.715994834899902,
                  'val/DER': 6.557376861572266,
                  'val/SER': 18.896764755249023}],
 'test_metrics': [{'test/loss': 0.9963287711143494,
                   'test/CER': 31.294574737548828,
                   'test/IER': 5.575967311859131,
                   'test/DER': 2.766371250152588,
                   'test/SER': 22.95223617553711}],
 'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-11/rnn-resume/checkpoints/epoch=116-step=14040.ckpt'}
```

VS Code, 80 epochs:

```
-----  
Test metric      DataLoader 0  
-----  
test/CER         38.75080871582031  
test/DER         2.9824941158294678  
test/IER         10.330667495727539  
test/SER         25.43764877319336  
test/loss        1.2315398454666138  
-----  
{'val_metrics': [{'val/loss': 1.3661220073699951,  
                    'val/CER': 41.82543182373047,  
                    'val/IER': 11.586175918579102,  
                    'val/DER': 7.864421844482422,  
                    'val/SER': 22.374834060668945}],  
  'test_metrics': [{'test/loss': 1.2315398454666138,  
                    'test/CER': 38.75080871582031,  
                    'test/IER': 10.330667495727539,  
                    'test/DER': 2.9824941158294678,  
                    'test/SER': 25.43764877319336}],  
  'best_checkpoint': 'C:\\\\Users\\Anna\\Desktop\\emg2qwerty\\logs\\  
2026-03-10\\new-rnn-run/checkpoints\\epoch=65-step=7920.ckpt'}
```

```
trainer.devices=1  
trainer.max_epochs=80  
batch_size=32  
num_workers=4  
module._target_=emg2qwerty.lightning.RNNCTCModule  
~module.block_channels  
~module.kernel_width  
+module.hidden_size=384  
+module.num_layers=3  
+module.dropout=0.2  
~cluster optimizer.lr=0.0001  
lr_scheduler.scheduler._target_=torch.optim.lr_scheduler.ConstantLR  
+lr_scheduler.scheduler.factor=1.0  
+lr_scheduler.scheduler.total_iters=80  
~lr_scheduler.scheduler.T_max  
~lr_scheduler.scheduler.eta_min  
hydra.run.dir=logs/2026-03-10/new-rnn-run
```

RNN is not doing well.

←  question @454



Vanilla RNN

Updated 5 days ago by Anonymous Comp

The accuracy is very low for the vanilla RNN. Are u guys getting the same result?

project

 Edit

 0



94 views

Students' Answer



Updated 5 days ago by Anonymous Atom

The professor suggested going straight to a LSTM or GRU model during lecture due to the issue of vanishing gradients.

 Edit

 1

Endorsed by Instructor (CHANG, YU-WEI)

1 Followup Discussions



Resolved

@454_f1



Anonymous Calc 5 days ago

Our group didn't test vanilla RNN at all.

 0

← Reply

Colab Run 2:

```
!python -m emg2qwerty.train \
  user="single_user" \
  dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \
  trainer.accelerator=gpu trainer.devices=1 \
  trainer.max_epochs=150 \
  module._target_=emg2qwerty.lightning.RNNCTCModule \
  ~module.block_channels ~module.kernel_width \
  +module.hidden_size=512 \
  +module.num_layers=4 \
  +module.dropout=0.3
```


Runningstage.validating	
metric	DataLoader 0
val/CER	50.221534729003906
val/DER	13.09260082244873
val/IER	12.53876781463623
val/SER	24.590164184570312
val/loss	1.5146290063858032

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
Testing DataLoader 0: 100% 1/1 [00:13<00:00, 13.71s/it]

Runningstage.testing	
metric	DataLoader 0
test/CER	38.923709869384766
test/DER	2.139615297317505
test/IER	12.059649467468262
test/SER	24.724443435668945
test/loss	1.2314406633377075

```
{'val_metrics': [{'val/loss': 1.5146290063858032,
                  'val/CER': 50.221534729003906,
                  'val/IER': 12.53876781463623,
                  'val/DER': 13.09260082244873,
                  'val/SER': 24.590164184570312}],
 'test_metrics': [{'test/loss': 1.2314406633377075,
                   'test/CER': 38.923709869384766,
                   'test/IER': 12.059649467468262,
                   'test/DER': 2.139615297317505,
                   'test/SER': 24.724443435668945}],
 'best_checkpoint':
 '/content/drive/MyDrive/emg2qwerty/logs/2026-03-12/02-45-01/checkpoints/epoch=147-step=17760.ckpt'
}
```

Runningstage.validating metric	DataLoader 0
val/CER	50.221534729003906
val/DER	13.09260082244873
val/IER	12.53876781463623
val/SER	24.590164184570312
val/loss	1.5146290063858032

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:13<00:00, 13.71s/it]

Runningstage.testing metric	DataLoader 0
test/CER	38.923709869384766
test/DER	2.139615297317505
test/IER	12.059649467468262
test/SER	24.724443435668945
test/loss	1.2314406633377075

```
{'val_metrics': [{'val/loss': 1.5146290063858032,
                  'val/CER': 50.221534729003906,
                  'val/IER': 12.53876781463623,
                  'val/DER': 13.09260082244873,
                  'val/SER': 24.590164184570312}],
 'test_metrics': [{'test/loss': 1.2314406633377075,
                   'test/CER': 38.923709869384766,
                   'test/IER': 12.059649467468262,
                   'test/DER': 2.139615297317505,
                   'test/SER': 24.724443435668945}],
 'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-12/02-45-01/checkpoints/epoch=147-step=17760.ckpt'}
```

	Name	Type	Params
0	model	Sequential	7.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
7.3 M	Trainable params		
0	Non-trainable params		
7.3 M	Total params		
29.235	Total estimated model params size (MB)		

to_tensor:

target: emg2qwerty.transforms.ToTensor

fields:

- emg_left
- emg_right

band_rotation:

target: emg2qwerty.transforms.ForEach

transform:

target: emg2qwerty.transforms.RandomBandRotation

offsets:

- -1
- 0

```

    - 1
temporal_jitter:
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter
  max_offset: 120
logspec:
  _target_: emg2qwerty.transforms.LogSpectrogram
  n_fft: 64
  hop_length: 16
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 3
  time_mask_param: 25
  n_freq_masks: 2
  freq_mask_param: 4
transforms:
  train:
    - ${to_tensor}
    - ${band_rotation}
    - ${temporal_jitter}
    - ${logspec}
    - ${specaug}
  val:
    - ${to_tensor}
    - ${logspec}
  test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.RNNCTCModule
  in_features: 528
  mlp_features:
    - 384
  hidden_size: 512
  num_layers: 4
  dropout: 0.3
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
  warmup_epochs: 10
  max_epochs: ${trainer.max_epochs}

```

```
warmup_start_lr: 1.0e-08
eta_min: 1.0e-06
interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150
```

Run with augmentation:

```
!python -m emg2qwerty.train \
  user="single_user" \
  dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \
  trainer.accelerator=gpu trainer.devices=1 \
  trainer.max_epochs=200 \
  module._target_=emg2qwerty.lightning.RNNCTCModule \
  ~module.block_channels ~module.kernel_width \
  +module.hidden_size=384 \
  +module.num_layers=3 \
  +module.dropout=0.2 \
  specaug.n_time_masks=1 \
  specaug.time_mask_param=10 \
  specaug.n_freq_masks=1 \
  specaug.freq_mask_param=2
```

```
to_tensor:
  _target_: emg2qwerty.transforms.ToTensor
fields:
  - emg_left
  - emg_right
band_rotation:
  _target_: emg2qwerty.transforms.ForEach
transform:
```

```

    _target_: emg2qwerty.transforms.RandomBandRotation
    offsets:
      - -1
      - 0
      - 1
temporal_jitter:
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter
  max_offset: 120
logspec:
  _target_: emg2qwerty.transforms.LogSpectrogram
  n_fft: 64
  hop_length: 16
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 1
  time_mask_param: 10
  n_freq_masks: 1
  freq_mask_param: 2
transforms:
  train:
    - ${to_tensor}
    - ${band_rotation}
    - ${temporal_jitter}
    - ${logspec}
    - ${specaug}
  val:
    - ${to_tensor}
    - ${logspec}
  test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.RNNCTCModule
  in_features: 528
  mlp_features:
    - 384
  hidden_size: 384
  num_layers: 3
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_: pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10

```

```

    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 200
  default_root_dir: ${hydra:runtime.output_dir}
callbacks:
- _target_: pytorch_lightning.callbacks.LearningRateMonitor
- _target_: pytorch_lightning.callbacks.ModelCheckpoint
  dirpath: ${hydra:runtime.output_dir}/checkpoints
  monitor: ${monitor_metric}
  mode: ${monitor_mode}
  save_last: true
  verbose: true

```

	Name	Type	Params
0	model	Sequential	3.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
3.7 M	Trainable params		
0	Non-trainable params		
3.7 M	Total params		
14.934	Total estimated model params size (MB)		

RNN with augmentation took 7 hours to train!

 Executing (6h 53m 34s) Connecting to T4 (Python 3)

Runningstage.validating metric	DataLoader 0
val/CER	40.56269454956055
val/DER	10.766504287719727
val/IER	9.392999649047852
val/SER	20.40319061279297
val/loss	1.270063042640686

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.77s/it]

Runningstage.testing metric	DataLoader 0
test/CER	33.239681243896484
test/DER	2.312513589859009
test/IER	8.212664604187012
test/SER	22.714502334594727
test/loss	1.0702632665634155

```
{'val_metrics': [{'val/loss': 1.270063042640686,
  'val/CER': 40.56269454956055,
  'val/IER': 9.392999649047852,
  'val/DER': 10.766504287719727,
  'val/SER': 20.40319061279297}],
'test_metrics': [{'test/loss': 1.0702632665634155,
  'test/CER': 33.239681243896484,
  'test/IER': 8.212664604187012,
  'test/DER': 2.312513589859009,
  'test/SER': 22.714502334594727}],
'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/19-31-04/checkpoints/epoch=195-step=23520.ckpt'}
```

Runningstage.validating metric	DataLoader 0
val/CER	40.56269454956055
val/DER	10.766504287719727
val/IER	9.392999649047852
val/SER	20.40319061279297
val/loss	1.270063042640686

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:11<00:00, 11.77s/it]

Runningstage.testing metric	DataLoader 0
test/CER	33.239681243896484
test/DER	2.312513589859009
test/IER	8.212664604187012
test/SER	22.714502334594727

```
| test/loss | 1.0702632665634155 |
{'val_metrics': [{'val/loss': 1.270063042640686,
                    'val/CER': 40.56269454956055,
                    'val/IER': 9.392999649047852,
                    'val/DER': 10.766504287719727,
                    'val/SER': 20.40319061279297}],
 'test_metrics': [{'test/loss': 1.0702632665634155,
                    'test/CER': 33.239681243896484,
                    'test/IER': 8.212664604187012,
                    'test/DER': 2.312513589859009,
                    'test/SER': 22.714502334594727}],
 'best_checkpoint':
 '/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/19-31-04/checkpoints/epoch=195-step=23520.ckpt'}
```

CNN + RNN

- Replaced TDSConvEncoder with CNNRNNEncoder
- Running on Kaggle.

	Run 1	Run 2	Run 3	Run 4
num_parameters	7.7M	10.5M	7.7 M	7.7 M
hidden_size	384	512	384	384
num_rnn_layers	2	3	2	2
dropout	0.2	0.3	0.3	0.4
scheduler	LinearWarmup CosineAnnealingLR	LinearWarmup CosineAnnealingLR	LinearWarmup CosineAnnealingLR	LinearWarmup CosineAnnealingLR
optimizer.lr	0.001	0.001	0.0005	0.001
optimizer.weight_decay				0.0001
warmup_epochs	10	10	10	10
max_epochs	100	150	200	100

specaug.n_time_masks	3	3	5	3
specaug.time_mask_param	25	25	50	25
val CER	17.28	16.86	17.90	17.811
test CER	19.60	20.19	20.01	20.27

Run 1:

```

  | Name      | Type          | Params
-----
0 | model      | Sequential    | 7.7 M
1 | ctc_loss   | CTCLoss      | 0
2 | metrics    | ModuleDict    | 0
-----
7.7 M      Trainable params
0          Non-trainable params
7.7 M      Total params
30.632     Total estimated model params size (MB)

```

module:

```

_target_: emg2qwerty.lightning.CNNRNNCTCModule
in_features: 528
mlp_features:
- 384
block_channels:
- 24
- 24
- 24
- 24
kernel_width: 32
hidden_size: 384
num_rnn_layers: 2
dropout: 0.2

```

datamodule:

```

_target_: emg2qwerty.lightning.WindowedEMGDataModule
window_length: 8000
padding:
- 1800
- 200

```

optimizer:

```

    _target_: torch.optim.Adam
    lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 100

```

	Name	Type	Params
0	model	Sequential	7.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

7.7 M	Trainable params
0	Non-trainable params
7.7 M	Total params
30.632	Total estimated model params size (MB)

Runningstage.validating metric	DataLoader 0
val/CER	17.27957534790039
val/DER	1.794417381286621
val/IER	3.278688430786133
val/SER	12.20646858215332
val/loss	0.8316115140914917

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0,1]

Testing DataLoader 0: 100%|██| 1/1 [00:09<00:00, 9.38s/it]

Runningstage.testing metric	DataLoader 0
test/CER	19.60233497619629
test/DER	2.5286362171173096
test/IER	2.63669753074646
test/SER	14.437000274658203
test/loss	0.8968871831893921

```
{'val_metrics': [{'val/loss': 0.8316115140914917,
                  'val/CER': 17.27957534790039,
                  'val/IER': 3.278688430786133,
                  'val/DER': 1.794417381286621,
                  'val/SER': 12.20646858215332}],
 'test_metrics': [{'test/loss': 0.8968871831893921,
                   'test/CER': 19.60233497619629,
                   'test/IER': 2.63669753074646,
                   'test/DER': 2.5286362171173096,
                   'test/SER': 14.437000274658203}],
 'best_checkpoint': '/kaggle/working/emg2qwerty/logs/2026-03-11/03-13-05/checkpoints/epoch=97-step=11760.ckpt'}
```

Run 2:

	Name	Type	Params
0	model	Sequential	10.5 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

10.5 M	Trainable params
0	Non-trainable params
10.5 M	Total params
42.179	Total estimated model params size (MB)

module:

```
_target_: emg2qwerty.lightning.CNNRRNNCTCModule
in_features: 528
mlp_features:
- 384
```

```

block_channels:
- 24
- 24
- 24
- 24
kernel_width: 32
hidden_size: 512
num_rnn_layers: 3
dropout: 0.3
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
  - 1800
  - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150

```

	Name	Type	Params
0	model	Sequential	10.5 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

10.5 M Trainable params
0 Non-trainable params
10.5 M Total params
42.179 Total estimated model params size (MB)

Runningstage.validating metric	DataLoader 0
val/CER	16.858661651611328
val/DER	2.0824103355407715
val/IER	2.813469171524048
val/SER	11.962782859802246
val/loss	0.7352327108383179

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0,1]
Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.97s/it]

Runningstage.testing metric	DataLoader 0
test/CER	20.18586540222168
test/DER	2.139615297317505
test/IER	2.593473196029663
test/SER	15.452776908874512
test/loss	0.8491751551628113

```
{'val_metrics': [{'val/loss': 0.7352327108383179,
                  'val/CER': 16.858661651611328,
                  'val/IER': 2.813469171524048,
                  'val/DER': 2.0824103355407715,
                  'val/SER': 11.962782859802246}],
 'test_metrics': [{'test/loss': 0.8491751551628113,
                   'test/CER': 20.18586540222168,
                   'test/IER': 2.593473196029663,
                   'test/DER': 2.139615297317505,
                   'test/SER': 15.452776908874512}],
 'best_checkpoint': '/kaggle/working/emg2qwerty/logs/2026-03-11/08-08-15/checkpoints/epoch=130-step=15720.ckpt'}
```

Run 3:

```
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 5
  time_mask_param: 50
  n_freq_masks: 2
  freq_mask_param: 4
module:
  _target_: emg2qwerty.lightning.CNNRRNNCTCModule
  in_features: 528
  mlp_features:
```

```

- 384
block_channels:
- 24
- 24
- 24
- 24
kernel_width: 32
hidden_size: 384
num_rnn_layers: 2
dropout: 0.3
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
  - 1800
  - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.0005
lr_scheduler:
  scheduler:
    _target_: pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 200

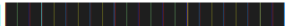
```

	Name	Type	Params
0	model	Sequential	7.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
7.7 M	Trainable params		
0	Non-trainable params		

7.7 M Total params
30.632 Total estimated model params size (MB)

Runningstage.validating metric	DataLoader 0
val/CER	17.89986801147461
val/DER	1.9716438055038452
val/IER	3.278688430786133
val/SER	12.649535179138184
val/loss	0.9750535488128662

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0,1]

Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.30s/it]

Runningstage.testing metric	DataLoader 0
test/CER	20.012968063354492
test/DER	1.9667171239852905
test/IER	2.9176571369171143
test/SER	15.128593444824219
test/loss	1.0850898027420044

```
{'val_metrics': [{'val/loss': 0.9750535488128662,
                  'val/CER': 17.89986801147461,
                  'val/IER': 3.278688430786133,
                  'val/DER': 1.9716438055038452,
                  'val/SER': 12.649535179138184}],
 'test_metrics': [{'test/loss': 1.0850898027420044,
                   'test/CER': 20.012968063354492,
                   'test/IER': 2.9176571369171143,
                   'test/DER': 1.9667171239852905,
                   'test/SER': 15.128593444824219}],
 'best_checkpoint': '/kaggle/working/emg2qwerty/logs/2026-03-12/05-17-13/checkpoints/epoch=169-step=20400.ckpt'}
```

Run 4:

```
!python -m emg2qwerty.train \
  user="single_user" \
  dataset.root=./data/89335547 \
  trainer.accelerator=gpu trainer.devices=1 \
  trainer.max_epochs=100 \
  module._target_=emg2qwerty.lightning.CNNRRNNCTCModule \
  +module.hidden_size=384 \
  +module.num_rnn_layers=2 \
  +module.dropout=0.4 \
  +optimizer.weight_decay=0.0001
```

```
module:
  _target_: emg2qwerty.lightning.CNNRRNNCTCModule
  in_features: 528
```

```

mlp_features:
- 384
block_channels:
- 24
- 24
- 24
- 24
kernel_width: 32
hidden_size: 384
num_rnn_layers: 2
dropout: 0.4
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
  - 1800
  - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
  weight_decay: 0.0001
lr_scheduler:
  scheduler:
    _target_: pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
  interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 100

```

	Name	Type	Params
0	model	Sequential	7.7 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

7.7 M Trainable params
0 Non-trainable params
7.7 M Total params
30.632 Total estimated model params size (MB)

Runningstage.validating metric	DataLoader 0
val/CER	17.811254501342773
val/DER	1.9494904279708862
val/IER	3.0793087482452393
val/SER	12.782454490661621
val/loss	0.8611534237861633

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0,1]
Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.34s/it]

Runningstage.testing metric	DataLoader 0
test/CER	20.272314071655273
test/DER	2.463799476623535
test/IER	2.463799476623535
test/SER	15.34471607208252
test/loss	0.9748815298080444

```
{'val_metrics': [{'val/loss': 0.8611534237861633,
                  'val/CER': 17.811254501342773,
                  'val/IER': 3.0793087482452393,
                  'val/DER': 1.9494904279708862,
                  'val/SER': 12.782454490661621}],
 'test_metrics': [{'test/loss': 0.9748815298080444,
                   'test/CER': 20.272314071655273,
                   'test/IER': 2.463799476623535,
                   'test/DER': 2.463799476623535,
                   'test/SER': 15.34471607208252}],
 'best_checkpoint': '/kaggle/working/emg2qwerty/logs/2026-03-12/10-21-55/checkpoints/epoch=89-step=10800.ckpt'}
```

Runningstage.validating metric	DataLoader 0
val/CER	17.811254501342773
val/DER	1.9494904279708862
val/IER	3.0793087482452393
val/SER	12.782454490661621
val/loss	0.8611534237861633

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0,1]
Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.34s/it]

Runningstage.testing	
metric	DataLoader 0
test/CER	20.272314071655273
test/DER	2.463799476623535
test/IER	2.463799476623535
test/SER	15.34471607208252
test/loss	0.9748815298080444

```
{'val_metrics': [{'val/loss': 0.8611534237861633,
                  'val/CER': 17.811254501342773,
                  'val/IER': 3.0793087482452393,
                  'val/DER': 1.9494904279708862,
                  'val/SER': 12.782454490661621}],
 'test_metrics': [{'test/loss': 0.9748815298080444,
                   'test/CER': 20.272314071655273,
                   'test/IER': 2.463799476623535,
                   'test/DER': 2.463799476623535,
                   'test/SER': 15.34471607208252}],
 'best_checkpoint':
 '/kaggle/working/emg2qwerty/logs/2026-03-12/10-21-55/checkpoints/epoch=89-step=10800.ckpt'}
```

CNN + GRU

- Replaced TDSConvEncoder with CNNGRUEncoder
- Running on Colab.

	Run 1	Run 2	Run 3	Run 4
num_parameters	11.2 M	11.3 M	11.3 M	11.3 M
hidden_size	384	384	384	384
num_rnn_layers	2	2	2	2
dropout	0.2	0.2	0.2	0.2
scheduler	LinearWarmupCosineAnnealingLR	LinearWarmupCosineAnnealingLR	LinearWarmupCosineAnnealingLR	LinearWarmupCosineAnnealingLR

warmup_epochs	10	10	10	10
optimizer.lr	0.001	0.001	0.001	0.001
max_epochs	150	150	150	150
block_channels	[24,24,24,24]	[32,32,32,32]	[32,32,32,32]	[32,32,32,32]
specaug.n_time_masks	3	3	0	1
specaug.time_mask_param	25	25	—	10
specaug.n_freq_masks	2	2	0	1
specaug.freq_mask_param	4	4	—	2
val CER	15.42	15.46	14.16	14.42
test CER	15.65	15.28	15.06	14.63

On Google Colab

Run 1

```
!python -m emg2qwerty.train \
  user="single_user" \
  dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \
  trainer.accelerator=gpu trainer.devices=1 \
  trainer.max_epochs=150 \
  module._target_=emg2qwerty.lightning.CNNGRUCTCModule \
  +module.hidden_size=384 \
  +module.num_rnn_layers=2 \
  +module.dropout=0.2
```

	Name	Type	Params
0	model	Sequential	11.2 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

11.2 M	Trainable params		
0	Non-trainable params		
11.2 M	Total params		
44.812	Total estimated model params size (MB)		

```

to_tensor:
  _target_: emg2qwerty.transforms.ToTensor
  fields:
    - emg_left
    - emg_right
band_rotation:
  _target_: emg2qwerty.transforms.ForEach
  transform:
    _target_: emg2qwerty.transforms.RandomBandRotation
    offsets:
      - -1
      - 0
      - 1
temporal_jitter:
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter
  max_offset: 120
logspec:
  _target_: emg2qwerty.transforms.LogSpectrogram
  n_fft: 64
  hop_length: 16
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 3
  time_mask_param: 25
  n_freq_masks: 2
  freq_mask_param: 4
transforms:
  train:
    - ${to_tensor}
    - ${band_rotation}
    - ${temporal_jitter}
    - ${logspec}
    - ${specaug}
  val:
    - ${to_tensor}
    - ${logspec}

```

```
    test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.CNNGRUCTCModule
  in_features: 528
  mlp_features:
    - 384
  block_channels:
    - 24
    - 24
    - 24
    - 24
  kernel_width: 32
  hidden_size: 384
  num_rnn_layers: 2
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150
  default_root_dir: ${hydra:runtime.output_dir}
```

```

callbacks:
- _target_: pytorch_lightning.callbacks.LearningRateMonitor
- _target_: pytorch_lightning.callbacks.ModelCheckpoint
  dirpath: ${hydra:runtime.output_dir}/checkpoints
  monitor: ${monitor_metric}
  mode: ${monitor_mode}
  save_last: true
  verbose: true

```

	Name	Type	Params
0	model	Sequential	11.2 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0


```

11.2 M    Trainable params
0         Non-trainable params
11.2 M    Total params
44.812    Total estimated model params size (MB)

```

Runningstage.validating		
metric	DataLoader 0	
val/CER	15.418697357177734	
val/DER	1.4621179103851318	
val/IER	3.256535291671753	
val/SER	10.700044631958008	
val/loss	0.785735011100769	

```

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
Testing DataLoader 0: 100% 1/1 [00:08<00:00, 8.98s/it]

```

Runningstage.testing		
metric	DataLoader 0	
test/CER	15.647287368774414	
test/DER	1.404797911643982	
test/IER	3.0257186889648438	
test/SER	11.216771125793457	
test/loss	0.7556009292602539	

```

{'val_metrics': [{'val/loss': 0.785735011100769,

```

```

        'val/CER': 15.418697357177734,
        'val/IER': 3.256535291671753,
        'val/DER': 1.4621179103851318,
        'val/SER': 10.700044631958008}],
    'test_metrics': [{'test/loss': 0.7556009292602539,
        'test/CER': 15.647287368774414,
        'test/IER': 3.0257186889648438,
        'test/DER': 1.404797911643982,
        'test/SER': 11.216771125793457}],
    'best_checkpoint': ''}

```

Runningstage.validating metric	DataLoader 0
val/CER	15.418697357177734
val/DER	1.4621179103851318
val/IER	3.256535291671753
val/SER	10.700044631958008
val/loss	0.785735011100769

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
 Testing DataLoader 0: 100% 1/1 [00:08<00:00, 8.98s/it]

Runningstage.testing metric	DataLoader 0
test/CER	15.647287368774414
test/DER	1.404797911643982
test/IER	3.0257186889648438
test/SER	11.216771125793457
test/loss	0.7556009292602539

```

{'val_metrics': [{'val/loss': 0.785735011100769,
    'val/CER': 15.418697357177734,
    'val/IER': 3.256535291671753,
    'val/DER': 1.4621179103851318,
    'val/SER': 10.700044631958008}],
    'test_metrics': [{'test/loss': 0.7556009292602539,
        'test/CER': 15.647287368774414,
        'test/IER': 3.0257186889648438,
        'test/DER': 1.404797911643982,
        'test/SER': 11.216771125793457}],
    'best_checkpoint': ''}

```

On Google Colab

Run 2:

```

!python -m emg2qwerty.train \
    user="single_user" \
    dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \

```

```
trainer.accelerator=gpu trainer.devices=1 \
trainer.max_epochs=150 \
module._target_=emg2qwerty.lightning.CNNGRUCTCModule \
+module.block_channels=[32,32,32,32] \ # increase channel width
+module.hidden_size=384 \
+module.num_rnn_layers=2 \
+module.dropout=0.2
```

Runningstage.validating metric	DataLoader 0
val/CER	15.463004112243652
val/DER	1.8830305337905884
val/IER	3.212228536605835
val/SER	10.367744445800781
val/loss	0.7308160066604614

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.67s/it]

Runningstage.testing metric	DataLoader 0
test/CER	15.279878616333008
test/DER	1.1238383054733276
test/IER	3.3715150356292725
test/SER	10.784525871276855
test/loss	0.6987296342849731

```
{'val_metrics': [{'val/loss': 0.7308160066604614,
                  'val/CER': 15.463004112243652,
                  'val/IER': 3.212228536605835,
                  'val/DER': 1.8830305337905884,
                  'val/SER': 10.367744445800781}],
 'test_metrics': [{'test/loss': 0.6987296342849731,
                   'test/CER': 15.279878616333008,
                   'test/IER': 3.3715150356292725,
                   'test/DER': 1.1238383054733276,
                   'test/SER': 10.784525871276855}],
 'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-12/19-35-45/checkpoints/epoch=127-step=15360.ckpt'}
```

Runningstage.validating metric	DataLoader 0
val/CER	15.463004112243652
val/DER	1.8830305337905884
val/IER	3.212228536605835
val/SER	10.367744445800781
val/loss	0.7308160066604614

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]
Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.67s/it]

Runningstage.testing metric	DataLoader 0
test/CER	15.279878616333008
test/DER	1.1238383054733276
test/IER	3.3715150356292725
test/SER	10.784525871276855
test/loss	0.6987296342849731

```
{'val_metrics': [{'val/loss': 0.7308160066604614,
                    'val/CER': 15.463004112243652,
                    'val/IER': 3.212228536605835,
                    'val/DER': 1.8830305337905884,
                    'val/SER': 10.367744445800781}],
 'test_metrics': [{'test/loss': 0.6987296342849731,
                    'test/CER': 15.279878616333008,
                    'test/IER': 3.3715150356292725,
                    'test/DER': 1.1238383054733276,
                    'test/SER': 10.784525871276855}],
 'best_checkpoint':
 '/content/drive/MyDrive/emg2qwerty/logs/2026-03-12/19-35-45/checkpoints/epoch=127-step=15360.ckpt'}
```

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

```
to_tensor:
  _target_: emg2qwerty.transforms.ToTensor
fields:
- emg_left
- emg_right
band_rotation:
  _target_: emg2qwerty.transforms.ForEach
transform:
  _target_: emg2qwerty.transforms.RandomBandRotation
```

```

        offsets:
        - -1
        - 0
        - 1
temporal_jitter:
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter
  max_offset: 120
logspec:
  _target_: emg2qwerty.transforms.LogSpectrogram
  n_fft: 64
  hop_length: 16
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 3
  time_mask_param: 25
  n_freq_masks: 2
  freq_mask_param: 4
transforms:
  train:
  - ${to_tensor}
  - ${band_rotation}
  - ${temporal_jitter}
  - ${logspec}
  - ${specaug}
  val:
  - ${to_tensor}
  - ${logspec}
  test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.CNNGRUCTCModule
  in_features: 528
  mlp_features:
  - 384
  block_channels:
  - 32
  - 32
  - 32
  - 32
  kernel_width: 32
  hidden_size: 384
  num_rnn_layers: 2
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
  - 1800
  - 200

```

```

optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150
  default_root_dir: ${hydra:runtime.output_dir}
callbacks:
- _target_: pytorch_lightning.callbacks.LearningRateMonitor
- _target_: pytorch_lightning.callbacks.ModelCheckpoint
  dirpath: ${hydra:runtime.output_dir}/checkpoints
  monitor: ${monitor_metric}
  mode: ${monitor_mode}
  save_last: true
  verbose: true

```

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

Run 3:

With data augmentation:

Removing SpecAugment:

```
!python -m emg2qwerty.train \  
  user="single_user" \  
  dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \  
  trainer.accelerator=gpu trainer.devices=1 \  
  trainer.max_epochs=150 \  
  module._target_=emg2qwerty.lightning.CNNGRUCTCModule \  
  +module.block_channels=[32,32,32,32] \  
  +module.hidden_size=384 \  
  +module.num_rnn_layers=2 \  
  +module.dropout=0.2 \  
  specaug.n_time_masks=0 \  
  specaug.n_freq_masks=0
```

```
to_tensor:  
  _target_: emg2qwerty.transforms.ToTensor  
  fields:  
    - emg_left  
    - emg_right  
band_rotation:  
  _target_: emg2qwerty.transforms.ForEach  
  transform:  
    _target_: emg2qwerty.transforms.RandomBandRotation  
    offsets:  
      - -1  
      - 0  
      - 1  
temporal_jitter:  
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter  
  max_offset: 120  
logspec:  
  _target_: emg2qwerty.transforms.LogSpectrogram  
  n_fft: 64  
  hop_length: 16  
specaug:  
  _target_: emg2qwerty.transforms.SpecAugment  
  n_time_masks: 0  
  time_mask_param: 25  
  n_freq_masks: 0  
  freq_mask_param: 4  
transforms:  
  train:
```

```

- ${to_tensor}
- ${band_rotation}
- ${temporal_jitter}
- ${logspec}
- ${specaug}
val:
- ${to_tensor}
- ${logspec}
test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.CNNGRUCTCModule
  in_features: 528
  mlp_features:
    - 384
  block_channels:
    - 32
    - 32
    - 32
    - 32
  kernel_width: 32
  hidden_size: 384
  num_rnn_layers: 2
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800
    - 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
    warmup_epochs: 10
    max_epochs: ${trainer.max_epochs}
    warmup_start_lr: 1.0e-08
    eta_min: 1.0e-06
    interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null

```

```

monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150
  default_root_dir: ${hydra:runtime.output_dir}
callbacks:
- _target_: pytorch_lightning.callbacks.LearningRateMonitor
- _target_: pytorch_lightning.callbacks.ModelCheckpoint
  dirpath: ${hydra:runtime.output_dir}/checkpoints
  monitor: ${monitor_metric}
  mode: ${monitor_mode}
  save_last: true
  verbose: true

```

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

Runningstage.validating metric	DataLoader 0
val/CER	14.155959129333496
val/DER	1.5064244270324707
val/IER	3.322995185852051
val/SER	9.326539993286133
val/loss	0.7045055627822876

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.59s/it]

Runningstage.testing metric	DataLoader 0
test/CER	15.063755989074707
test/DER	1.0590015649795532
test/IER	3.393127202987671
test/SER	10.611627578735352
test/loss	0.7594947218894958

```
{'val_metrics': [{'val/loss': 0.7045055627822876,
  'val/CER': 14.155959129333496,
  'val/IER': 3.322995185852051,
  'val/DER': 1.5064244270324707,
  'val/SER': 9.326539993286133}],
'test_metrics': [{'test/loss': 0.7594947218894958,
  'test/CER': 15.063755989074707,
  'test/IER': 3.393127202987671,
  'test/DER': 1.0590015649795532,
  'test/SER': 10.611627578735352}],
'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/04-42-30/checkpoints/epoch=131-step=15840.ckpt'}
```

Runningstage.validating metric	DataLoader 0
val/CER	14.155959129333496
val/DER	1.5064244270324707
val/IER	3.322995185852051
val/SER	9.326539993286133
val/loss	0.7045055627822876

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:09<00:00, 9.59s/it]

Runningstage.testing metric	DataLoader 0
test/CER	15.063755989074707
test/DER	1.0590015649795532
test/IER	3.393127202987671

test/SER	10.611627578735352
test/loss	0.7594947218894958

```
{'val_metrics': [{'val/loss': 0.7045055627822876,
                  'val/CER': 14.155959129333496,
                  'val/IER': 3.322995185852051,
                  'val/DER': 1.5064244270324707,
                  'val/SER': 9.326539993286133}],
 'test_metrics': [{'test/loss': 0.7594947218894958,
                   'test/CER': 15.063755989074707,
                   'test/IER': 3.393127202987671,
                   'test/DER': 1.0590015649795532,
                   'test/SER': 10.611627578735352}],
 'best_checkpoint':
 '/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/04-42-30/checkpoints/epoch=131-step=15840.ckpt'}
```

Run 4:

Light augmentation

```
!python -m emg2qwerty.train \
  user="single_user" \
  dataset.root=/content/drive/MyDrive/emg2qwerty/data/89335547 \
  trainer.accelerator=gpu trainer.devices=1 \
  trainer.max_epochs=150 \
  module._target_=emg2qwerty.lightning.CNNGRUCTCModule \
  +module.block_channels=[32,32,32,32] \
  +module.hidden_size=384 \
  +module.num_rnn_layers=2 \
  +module.dropout=0.2 \
  specaug.n_time_masks=1 \
  specaug.time_mask_param=10 \
  specaug.n_freq_masks=1 \
  specaug.freq_mask_param=2
```

```
to_tensor:
  _target_: emg2qwerty.transforms.ToTensor
fields:
  - emg_left
  - emg_right
band_rotation:
  _target_: emg2qwerty.transforms.ForEach
transform:
```



```

    _target_: emg2qwerty.transforms.RandomBandRotation
    offsets:
      - -1
      - 0
      - 1
temporal_jitter:
  _target_: emg2qwerty.transforms.TemporalAlignmentJitter
  max_offset: 120
logspec:
  _target_: emg2qwerty.transforms.LogSpectrogram
  n_fft: 64
  hop_length: 16
specaug:
  _target_: emg2qwerty.transforms.SpecAugment
  n_time_masks: 1
  time_mask_param: 10
  n_freq_masks: 1
  freq_mask_param: 2
transforms:
  train:
    - ${to_tensor}
    - ${band_rotation}
    - ${temporal_jitter}
    - ${logspec}
    - ${specaug}
  val:
    - ${to_tensor}
    - ${logspec}
  test: ${transforms.val}
module:
  _target_: emg2qwerty.lightning.CNNGRUCTCModule
  in_features: 528
  mlp_features:
    - 384
  block_channels:
    - 32
    - 32
    - 32
    - 32
  kernel_width: 32
  hidden_size: 384
  num_rnn_layers: 2
  dropout: 0.2
datamodule:
  _target_: emg2qwerty.lightning.WindowedEMGDataModule
  window_length: 8000
  padding:
    - 1800

```

```

- 200
optimizer:
  _target_: torch.optim.Adam
  lr: 0.001
lr_scheduler:
  scheduler:
    _target_:
pl_bolts.optimizers.lr_scheduler.LinearWarmupCosineAnnealingLR
  warmup_epochs: 10
  max_epochs: ${trainer.max_epochs}
  warmup_start_lr: 1.0e-08
  eta_min: 1.0e-06
  interval: epoch
decoder:
  _target_: emg2qwerty.decoder.CTCGreedyDecoder
seed: 1501
batch_size: 32
num_workers: 4
train: true
checkpoint: null
monitor_metric: val/CER
monitor_mode: min
trainer:
  accelerator: gpu
  devices: 1
  num_nodes: 1
  max_epochs: 150
  default_root_dir: ${hydra:runtime.output_dir}
callbacks:
- _target_: pytorch_lightning.callbacks.LearningRateMonitor
- _target_: pytorch_lightning.callbacks.ModelCheckpoint
  dirpath: ${hydra:runtime.output_dir}/checkpoints
  monitor: ${monitor_metric}
  mode: ${monitor_mode}
  save_last: true
  verbose: true

```

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0
11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

	Name	Type	Params
0	model	Sequential	11.3 M
1	ctc_loss	CTCLoss	0
2	metrics	ModuleDict	0

11.3 M	Trainable params		
0	Non-trainable params		
11.3 M	Total params		
45.042	Total estimated model params size (MB)		

Runningstage.validating			
metric		DataLoader 0	

val/CER		14.421798706054688	
val/DER		1.4399645328521729	
val/IER		3.167922019958496	
val/SER		9.813912391662598	
val/loss		0.7255727052688599	

LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]

Testing DataLoader 0: 100% 1/1 [00:10<00:00, 10.13s/it]

Runningstage.testing			
metric		DataLoader 0	

test/CER		14.631510734558105	
test/DER		0.9293278455734253	
test/IER		2.6583099365234375	
test/SER		11.043872833251953	
test/loss		0.6987120509147644	

```
{'val_metrics': [{'val/loss': 0.7255727052688599,
                  'val/CER': 14.421798706054688,
                  'val/IER': 3.167922019958496,
                  'val/DER': 1.4399645328521729,
                  'val/SER': 9.813912391662598}],
 'test_metrics': [{'test/loss': 0.6987120509147644,
                   'test/CER': 14.631510734558105,
                   'test/IER': 2.6583099365234375,
                   'test/DER': 0.9293278455734253,
                   'test/SER': 11.043872833251953}]},
```

```
'best_checkpoint':  
'/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/09-27-58/checkpoints/epoch=136-step=16440.ckpt'}
```

Runningstage.validating metric	DataLoader 0
val/CER	14.421798706054688
val/DER	1.4399645328521729
val/IER	3.167922019958496
val/SER	9.813912391662598
val/loss	0.7255727052688599

```
LOCAL_RANK: 0 - CUDA_VISIBLE_DEVICES: [0]  
Testing DataLoader 0: 100% 1/1 [00:10<00:00, 10.13s/it]
```

Runningstage.testing metric	DataLoader 0
test/CER	14.631510734558105
test/DER	0.9293278455734253
test/IER	2.6583099365234375
test/SER	11.043872833251953
test/loss	0.6987120509147644

```
{'val_metrics': [{'val/loss': 0.7255727052688599,  
  'val/CER': 14.421798706054688,  
  'val/IER': 3.167922019958496,  
  'val/DER': 1.4399645328521729,  
  'val/SER': 9.813912391662598}],  
'test_metrics': [{'test/loss': 0.6987120509147644,  
  'test/CER': 14.631510734558105,  
  'test/IER': 2.6583099365234375,  
  'test/DER': 0.9293278455734253,  
  'test/SER': 11.043872833251953}],  
'best_checkpoint': '/content/drive/MyDrive/emg2qwerty/logs/2026-03-13/09-27-58/checkpoints/epoch=136-step=16440.ckpt'}
```

Experimenting with data augmentation:
Augmentation ablation: default (heavy) → none → light